

STABLE LOCAL ARITHMETIC FINGERPRINTS FROM LOCALIZED WEIL FORMS

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. To a localization of the Riemann–Weil explicit formula—a height T_0 , a width σ , and a finite basis of band-limited test functions—and to any L-function L (with Euler product, Γ -factor, and functional equation) we associate a finite Hermitian *localized Weil form* $Q(L)$, its bottom eigenvalue $\lambda_{\min}(Q(L))$, and a per-prime *anatomy profile* $\mathcal{R}(L) = (R_p(L))_p$. We prove that these objects are *exponentially stable local-arithmetic fingerprints*: if two L-functions share their local data $\Lambda_L(n)$ for all $n \leq P$, then

$$\|Q(L) - Q(L')\| \leq A e^{-\alpha(\log P)^2}, \quad |\lambda_{\min}(L) - \lambda_{\min}(L')| \leq A e^{-\alpha(\log P)^2},$$

with $\alpha \approx \sigma^2/8$ (unconditionally, from the Ramanujan bound and the Gaussian window), and, when the bottom eigenvalue is simple with spectral gap g , $\|\mathcal{R}(L) - \mathcal{R}(L')\| \leq (C/g) e^{-\alpha(\log P)^2}$. We further give an exact *anatomy identity* $\lambda_{\min} = R_{\infty} - \sum_p R_p$ realizing the bottom as a mass competition between an archimedean term and a per-prime prime-mass, an explicit Hellmann–Feynman gradient, and a *concavity + monotonicity* structure yielding an extremal reduction of positivity over coefficient boxes. All results are elementary and **independent of the Riemann Hypothesis**; they isolate, as a self-contained statement, the local-arithmetic robustness underlying localized explicit-formula detectors.

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1. INTRODUCTION

The Riemann–Weil explicit formula reads the positivity of zeta (and L-function) zeros as the positivity of a quadratic functional—Weil’s criterion. *Localized* versions of this functional, assembled on a finite basis of test functions concentrated near a height T_0 , have been used as numerical and rigorous probes of off-critical zeros [1]. This note isolates a structural fact about such localizations that is of independent interest and *does not depend on the Riemann Hypothesis*: the localized form, its bottom eigenvalue, and its per-prime profile are **exponentially stable functionals of the small-prime arithmetic** of the underlying L-function. In other words, the localized Weil form is a *local-arithmetic fingerprint*, robust to the (globally inaccessible) tail of the coefficients.

The point is simple but, we believe, worth recording cleanly: a Gaussian (or prolate) window of width σ in the spectral variable suppresses the contribution of a prime p super-exponentially in $\log p$, so the entire localized object is determined—up to error $e^{-\alpha(\log P)^2}$ —by the primes $p \leq P$. We make this a theorem (§4), and we accompany it with three elementary structural results that together give the fingerprint a usable internal geometry: an exact per-prime decomposition (§3), its Hellmann–Feynman gradient (§3), and a concavity/monotonicity extremal reduction (§5). What is and is not claimed. Nothing here proves, or attempts to prove, the Riemann Hypothesis. The bottom eigenvalue’s *sign* (local Weil positivity) is exactly the hard arithmetic question and is untouched; what we establish is the *stability and internal structure* of the fingerprint, which hold unconditionally.

2. SETUP

Fix $d > \frac{1}{2}$, a height T_0 , a width σ , and an orthonormal basis $\{\phi_1, \dots, \phi_J\}$ of band-limited test functions concentrated near T_0 in the spectral variable (e.g. Hermite–Gauss or Slepian prolates). For an L-function L with $-L'/L(s) = \sum_n \Lambda_L(n)n^{-s}$, completed Λ -factor and conductor N , the *localized Weil form* is the $J \times J$ Hermitian Gram matrix of the Weil functional $\mathfrak{t}_L$ on the basis, written via the explicit formula’s arithmetic side (prime cutoff X):

$$(1) \quad Q(L) := A_\infty - \sum_{n \leq X} \frac{\Lambda_L(n)}{\sqrt{n}} w_n w_n^*, \quad (w_n)_i := \widehat{\phi}_i(\log n),$$

where $A_\infty = A_\infty(\Gamma_L, N) \geq 0$ is the (L-independent within a family) archimedean+polar block, and $w_n \in \mathbb{C}^J$ is the basis’ frequency content at $\log n$. The only L-dependence is the *local data* $D_X(L) := \{\Lambda_L(n)\}_{n \leq X}$. We write $\lambda_{\min}(L) := \lambda_{\min}(Q(L))$.

Definition 2.1 (Anatomy profile). Let u_0 be a unit bottom eigenvector of $Q(L)$ and $\widehat{u}_0(x) := \sum_i (u_0)_i \widehat{\phi}_i(x)$. The *anatomy profile* is $\mathcal{R}(L) := (R_p(L))_p$ with

$$R_p(L) := \sum_{k \geq 1} \frac{\Lambda_L(p^k)}{\sqrt{p^k}} |\widehat{u}_0(k \log p)|^2.$$

We assume the standard **window decay** and **Ramanujan** inputs:

$$(2) \quad \|w_n\|^2 \leq W(\log n), \quad W(x) \leq A_0 e^{-\sigma^2 x^2/2} \quad (\text{Gaussian window}); \quad |\Lambda_L(p^k)| \leq (k+1) \log p.$$

(The Ramanujan bound holds for ζ and Dirichlet L trivially and for $\text{GL}(2)$ cusp forms by Deligne; for the stability theorem any polynomial bound on Λ_L suffices.)

3. THE ANATOMY IDENTITY AND GRADIENT

Proposition 3.1 (Anatomy identity). *With $R_\infty(u) := \langle u, A_\infty u \rangle$,*

$$\lambda_{\min}(L) = R_\infty(u_0) - \sum_p R_p(L) = R_\infty(u_0) - \|\mu_{u_0}\|,$$

where

$$\mu_u := \sum_{n \geq 2} \frac{\Lambda_L(n)}{\sqrt{n}} |\widehat{u}(\log n)|^2 \delta_{\log n}$$

is the anatomy measure. Equivalently $\lambda_{\min}(L) = \min_{\|u\|=1} [R_\infty(u) - \|\mu_u\|]$.

Proof. $\lambda_{\min} = \langle u_0, Qu_0 \rangle = \langle u_0, A_\infty u_0 \rangle - \sum_n \frac{\Lambda_L(n)}{\sqrt{n}} |\langle u_0, w_n \rangle|^2$; group $n = p^k$ and use $\langle u_0, w_{p^k} \rangle = \widehat{u_0}(k \log p)$. The variational form is the Rayleigh principle. $\square$

Proposition 3.2 (Gradient). *If $\lambda_{\min}(L)$ is simple with unit eigenvector u_0 , then for $n \leq X$*

$$\frac{\partial \lambda_{\min}}{\partial \Lambda_L(n)} = -\frac{1}{\sqrt{n}} |\widehat{u_0}(\log n)|^2 \leq 0.$$

Proof. $\partial Q / \partial \Lambda_L(n) = -\frac{1}{\sqrt{n}} w_n w_n^*$; Hellmann–Feynman for the simple eigenvalue $\lambda_{\min}$ gives

$$\partial \lambda_{\min} / \partial \Lambda_L(n) = \langle u_0, (\partial Q / \partial \Lambda_L(n)) u_0 \rangle.$$

$\square$

Thus the gradient is the (negative) density of μ_{u_0} : the anatomy profile is the first-order response of the bottom eigenvalue to the local arithmetic.

4. THE STABILITY THEOREM

Theorem 4.1 (Local-arithmetic stability). *Suppose $\Lambda_L(n) = \Lambda_{L'}(n)$ for all $n \leq P$, and both satisfy the inputs (2). Then, with $\alpha = \sigma^2/2$ and $A = A(\sigma, J, T_0)$:*

- (i) **(operator)** $\|Q(L) - Q(L')\| \leq A e^{-\alpha(\log P)^2}$;
- (ii) **(spectrum, unconditional)** $|\lambda_{\min}(L) - \lambda_{\min}(L')| \leq A e^{-\alpha(\log P)^2}$, and more generally $|\lambda_j(L) - \lambda_j(L')| \leq A e^{-\alpha(\log P)^2}$ for every eigenvalue;
- (iii) **(profile, gap-conditional)** if $\lambda_{\min}$ is simple with gap $g = \lambda_2 - \lambda_1 > 0$ on both, then $\|\mathcal{R}(L) - \mathcal{R}(L')\| \leq \frac{C}{g} e^{-\alpha(\log P)^2}$.

Proof. (i) Since the local data agree up to P , $Q(L) - Q(L') = -\sum_{n>P} \frac{\Lambda_L(n) - \Lambda_{L'}(n)}{\sqrt{n}} w_n w_n^*$, hence

$$\|Q(L) - Q(L')\| \leq \sum_{n>P} \frac{|\Lambda_L(n)| + |\Lambda_{L'}(n)|}{\sqrt{n}} \|w_n\|^2 \leq 2 \sum_{n>P} \frac{(\Omega(n) + 1) \log n}{\sqrt{n}} A_0 e^{-\sigma^2(\log n)^2/2},$$

using (2) ($\Omega(n)$ = number of prime factors with multiplicity; $|\Lambda_L(n)| \leq (\Omega(n) + 1) \log n$). The summand decays like $e^{-\sigma^2(\log n)^2/2}$, so the tail over $n > P$ is $\leq A e^{-\alpha(\log P)^2}$ with $\alpha = \sigma^2/2$ (absorbing the slowly varying factors into A and a harmless reduction of α). (ii) is Weyl's inequality applied to (i). (iii) By the Davis–Kahan $\sin \Theta$ theorem, the bottom eigenprojector obeys $\|P_1(L) - P_1(L')\| \leq \|Q(L) - Q(L')\|/g$; the unit eigenvector (up to phase) and hence $\widehat{u_0}$ and each R_p are Lipschitz in P_1 on the unit sphere, and $\sum_p$ of the window-weighted contributions converges by (2); combine with (i). $\square$

Corollary 4.2 (Fingerprint). *The map $L \mapsto (Q(L), \lambda_{\min}(L), \mathcal{R}(L))$ is, up to error $e^{-\alpha(\log P)^2}$, a function of the truncated local data $D_P(L) = \{\Lambda_L(n)\}_{n \leq P}$ alone. In particular it is a local invariant: two L -functions indistinguishable on small primes are indistinguishable, exponentially, by the localized Weil form.*

5. CONCAVITY, MONOTONICITY, AND AN EXTREMAL REDUCTION

Viewing $\lambda_{\min}$ as a function of the coefficient vector $\Lambda = (\Lambda_L(n))_{n \leq X}$:

Theorem 5.1 (Extremal reduction). (1) $\Lambda \mapsto \lambda_{\min}(Q(\Lambda))$ is **concave**.
 (2) On $\Lambda_n \geq 0$ it is **coordinate-wise non-increasing**.
 (3) On a box $\mathcal{B} = \{0 \leq \Lambda_n \leq c_n\}$, $\min_{\Lambda \in \mathcal{B}} \lambda_{\min}(Q(\Lambda)) = \lambda_{\min}(Q(c))$ is attained at the all-upper corner. Hence positivity ($\lambda_{\min} \geq 0$) holds throughout $\mathcal{B}$ iff it holds at the single corner $\Lambda \equiv c$.

Proof. (1) $\lambda_{\min}(M) = \min_{\|u\|=1} \langle u, Mu \rangle$ is concave in M ; $Q(\Lambda)$ is affine in Λ ; concave $\circ$ affine is concave. (2) $\partial Q / \partial \Lambda_n = -\frac{1}{\sqrt{n}} w_n w_n^* \preceq 0$ and $\lambda_{\min}(M - \text{PSD}) \leq \lambda_{\min}(M)$ (Weyl). (3) A concave function on a compact convex set attains its minimum at an extreme point; non-increasingness selects the upper corner. $\square$

Remark 5.2 (Reading). Theorem 5.1 reduces the positivity of an entire coefficient box to a single extremal object. It is a statement about the *prime masses* Λ_n (in which $\lambda_{\min}$ is concave and monotone), with no natural analogue in the raw normalized coefficients a_p (whose sign-oscillation hides the monotone structure). For the $\text{GL}(2)$ -tempered non-negative box $0 \leq a_{p^k} \leq 2$ the corner is the ζ^2 -point $a_{p^k} \equiv 2$; **the reduction is the theorem—whether the corner itself is positive is a separate question, not addressed here.**

6. RELATION TO LOCALIZED DETECTORS, AND SCOPE

Theorem 4.1 is the structural reason localized explicit-formula detectors [1] are well-defined and reproducible: the localized form depends on the arithmetic only through small primes, up to super-polynomially small error. Corollary 4.2 turns this into a classification-flavored statement—an exponentially stable local fingerprint—that stands on its own, independent of the Riemann Hypothesis. The *sign* of $\lambda_{\min}$ (local Weil positivity) is the hard arithmetic content and is deliberately left untouched: this paper concerns the fingerprint’s *stability and geometry*, not its sign.

REFERENCES

- [1] D. A. Trejo Pizzo, *A rigorous localized Weil detector for off-critical zeros*, 2026.
- [2] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, 1952.
- [3] T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1976.
- [4] C. Davis, W. M. Kahan, *The rotation of eigenvectors by a perturbation. III*, SIAM J. Numer. Anal. **7** (1970), 1–46.
- [5] P. Deligne, *La conjecture de Weil. I*, Publ. IHÉS **43** (1974), 273–307.
- [6] A. Connes and C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. (N.S.) **27** (2021), no. 77.
- [7] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloq. Publ. **53**, AMS, 2004.